

ORF309/MAT380

**Continuous Random Variables II:
Joint & Conditional Densities**

(pp. 70-74)

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Agenda for today

- Joint PDF for CRVs
- Conditional PDF for CRVs
- Example

Joint distribution of CRVs

- As in DRVs we need the **Joint Density (Joint PDF) of two CRVs** in order to compute their expectation
- If X, Y are two **DRVs** the expectation of $\varphi(X, Y)$ is defined

$$E[\varphi(X, Y)] = \sum_{x, y} \varphi(x, y) \underbrace{P\{X = x, Y = y\}}_{\text{Joint distribution of DRVs}}$$

If X, Y are two **CRVs** the **expectation** of $\varphi(X, Y)$ is defined

$$E[\varphi(X, Y)] = \int \varphi(x, y) \underbrace{P\{X \in dx, Y \in dy\}}_{\text{Joint distribution of CRVs}}$$

Joint distribution of CRVs

- Let's expand the expectation of $\varphi(X, Y)$

Derivations are similar to those in slide-deck 11

$$E[\varphi(X, Y)] = \int \varphi(x, y) P\{X \in dx, Y \in dy\} = \int \varphi(x, y) \frac{P\{X \in dx, Y \in dy\}}{dxdy} dxdy =$$

What does this remind you??



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Joint CDF: $F(x, y) = P\{X \leq x, Y \leq y\}$

Joint PDF: $f(x, y) = \frac{\partial^2}{\partial x \partial y} P\{X \leq x, Y \leq y\} = \frac{\partial^2}{\partial x \partial y} F(x, y)$

Marginal density

- Assume now that **we know** the **Joint density** $f(x, y)$ of two CRVs X, Y .
- How can we compute the **Marginal density of X** ??
 - Let's denote the marginal density of X as $f_X(x)$ (**unknown for now...**)
- We will use the expectation of $\varphi(X)$ and write it as $\varphi(X) = \psi(X, Y)$ for some function ψ :
 - ❖ Any ideas about how the function ψ could look like ??

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$$E[\varphi(X)] = E[\psi(X, Y)] = \int \int \psi(x, y) f(x, y) dx dy = \int \int \varphi(x) f(x, y) dy dx = \int \varphi(x) \underbrace{\int f(x, y) dy}_{f_X(x)} dx$$

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The Marginal density of X obtained from the Joint density of X and Y :

$$f_X(x) = \int f(x, y) dy$$

Additivity of expectation of two CRVs

- We know that for **any** two **DRVs** we have: $E[X + Y] = E[X] + E[Y]$
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$$= \int x \int f(x, y)dy dx + \int y \int f(x, y)dx dy = \int xf_X(x)dx + \int yf_Y(y)dy = E[X] + E[Y]$$

Conditional distributions for CRVs

- Conditional distribution for **DRVs**

$$P\{X = x|Y = y\} = \frac{P\{X = x, Y = y\}}{P\{Y = y\}}$$

- For **CRVs** we have $P\{Y = y\} = 0$ and hence the expression:

$$P\{X \in dx|Y = y\} = \frac{P\{X \in dx, Y = y\}}{P\{Y = y\}}$$

would have been **undefined!!**

Conditional distributions for CRVs

- But we can **replace** the event $\{Y = y\}$ **with** the event $\{y < Y \leq y + \Delta y\}$ for some **small** $\Delta y > 0$ and define the conditional distribution as the **limit**

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Conditional distributions for CRVs

- Expanding the previous definition further and considering the joint distribution $f(x, y)$ and the marginal distribution $f_Y(y)$ we obtain:

$$P\{X \in dx | Y = y\} = \frac{f(x, y)dx}{\left(\int f(x', y)dx'\right)} =$$

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Conditional distributions for CRVs

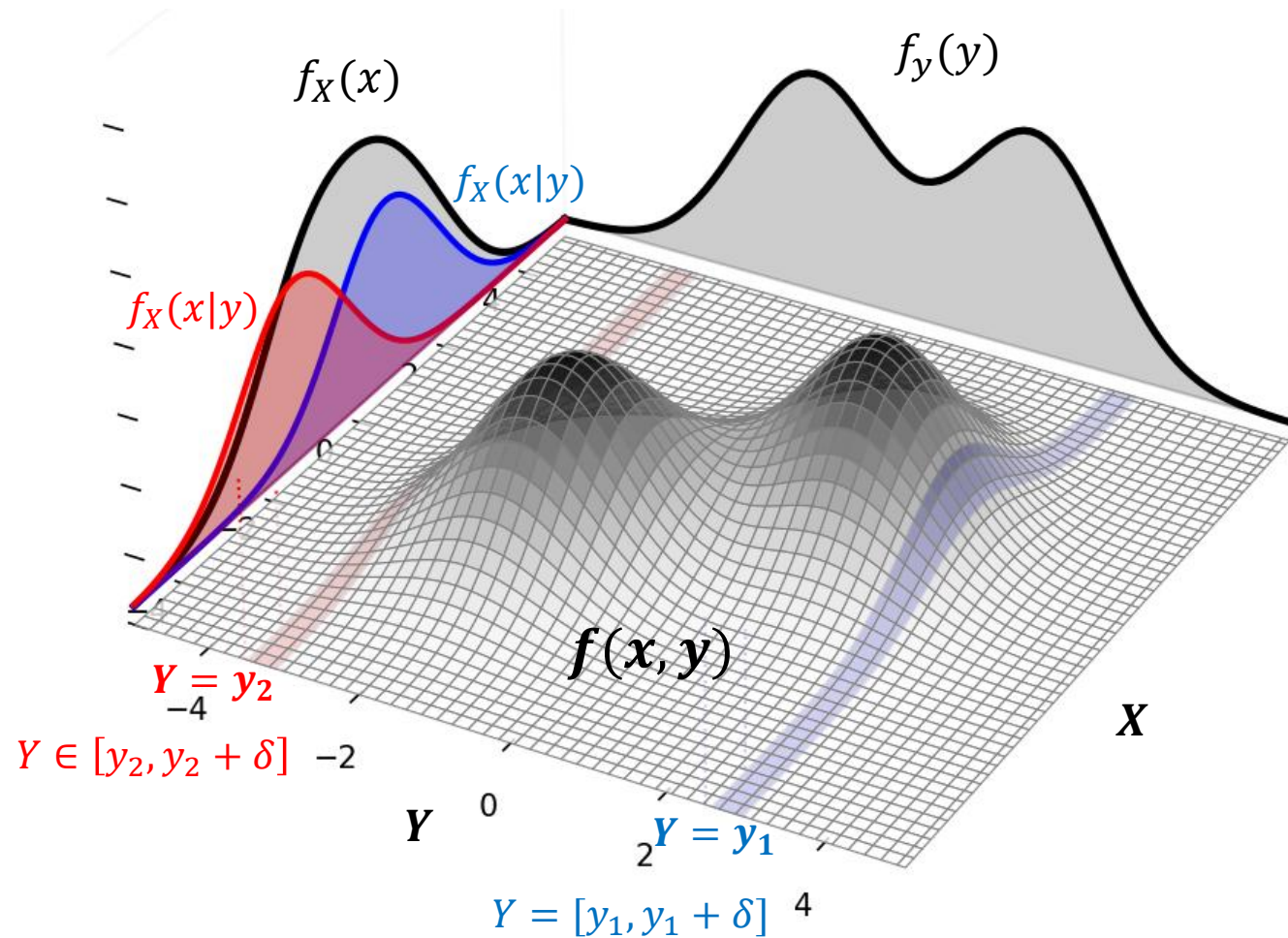
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- Hence the **conditional density of x given y** is defined as the function:

$$f(x|y) = \frac{f(x, y)}{\int f(x', y)dx'}$$

Visualization of conditional distributions



Conditional distributions for CRVs

- **Conditional expectation** of $\varphi(X)$ given $Y = y$ is the easy to define:

$$E[\varphi(X)|Y = y] = \int \varphi(X)P\{X \in dx|Y = y\} = \int \varphi(X)f(x|y)dx$$

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- **Unconditional Expectation** using Conditional Expectation (**Tower Property!**)

$$E[\varphi(X)] = \int \int \varphi(x)P\{X \in dx, Y \in dy\} = \int \int \varphi(x) \frac{P\{X \in dx, Y \in dy\}}{P\{Y \in dy\}} P\{y \in dy\} =$$

$$= \int \int \varphi(x)P\{X \in dx|Y = y\}P\{y \in dy\} = \int E[\varphi(X)|Y = y]P\{Y \in dy\}$$

Conditional distributions for CRVs

- **Conditional probability within an interval**
 - Use the expectation of the indicator RV within that interval

$$\begin{aligned} P\{a \leq X \leq b | Y = y\} &= E[\mathbf{1}_{a \leq X \leq b}] = \\ &= \int_{-\infty}^{\infty} \mathbf{1}_{a \leq x \leq b} P\{X \in [a, b] | Y = y\} dx = \int_a^b \frac{f(x, y)}{\int f(x', y) dx'} dx = \int_a^b f(x|y) dx \end{aligned}$$

Example

- **A guessing game:**
 - **Person 1** draws some random number from $X \sim \text{Expon}(\mu)$ distribution and shares its outcome $X = x$ with Person 2
 - **Person 2** draws a random number from $Y \sim \text{Expon}(x)$ and shares its outcome $Y = y$ with Person 3
 - **Person 3** needs to guess (or estimate) the number x that Person 1 drew
 - What is **our knowledge** so far?
 - $P\{X \in dx\} = \mu e^{-\mu x} dx$ and $P\{Y \in dy | X = x\} = x e^{-xy} dy$ (conditional on X)

Example

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$$E[X|Y = y] = \int x P\{X \in dx | Y = y\} = \int x \frac{P\{X \in dx, Y \in dy\}}{P\{y \in dy\}}$$

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Person 3 can compute the joint distribution of X, Y using the known conditional distribution

$$P\{Y \in dy | X = x\} = \frac{P\{X \in dx, Y \in dy\}}{P\{X \in dx\}} \Rightarrow$$

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$$P\{X \in dx, Y \in dy\} = P\{Y \in dy | X = x\} P\{X \in dx\} = \underbrace{xe^{-xy} dx}_{\text{Expon}(x)} \underbrace{\mu e^{-\mu x} dy}_{\text{Expon}(\mu)} = \mu x e^{-(\mu+y)x} dx dy$$

Example

- Person 3 can now also compute the **Marginal density** of Y

$$P\{Y \in dy\} = \left(\int_{-\infty}^{\infty} P\{X \in dx, Y \in dy\} dx \right) dy = \left(\int_0^{\infty} \mu x e^{-(\mu+y)x} dx \right) dy = \dots = \frac{\mu}{(\mu + y)^2} dy$$

- **Note:** You can compute the integral in two ways
 1. Using **Calculus** and **integration by parts** (which may take some time) OR
 2. Observing that the $\int_0^{\infty} x(\mu + y)e^{-(\mu+y)x} dx$ is the **expectation** of $\text{Expon}(\mu + y)$

Example

- Person 3 can compute now the conditional probability:

$$P\{X \in dx|Y = y\} = \frac{P\{X \in dx, Y \in dy\}}{P\{Y \in dy\}} = \frac{\mu x e^{-(\mu+y)x} dx dy}{\frac{\mu}{(\mu+y)^2} dy} = x(\mu+y)^2 e^{-(\mu+y)x} dx$$

- So, Person 3 has everything she/he needs to compute the **expectation**:

$$E[X|Y = y] = \int x P\{X \in dx|Y = y\} = \int x x(\mu+y)^2 e^{-(\mu+y)x} dx$$

Example

- The integral can be solved by using **Calculus** (*apply integration by parts twice*) or by recalling the **CDF** of the **Gamma**(2, $\mu + y$) distribution (*time of the 2nd arrival*)

$$P\{T \leq t\} = 1 - \sum_{n=0}^{2-1} \frac{e^{-(\mu+y)t} ((\mu+y)t)^n}{n!} = 1 - e^{-(\mu+y)t} - (\mu+y)t e^{-(\mu+y)t}, \quad \text{for } y \geq 0$$

and taking its **derivative**, we can find the **PDF** (which is the same as $P\{X \in dx | Y = y\}$)

$$P\{T \in dt\} = t(\mu+y)^2 e^{-(\mu+y)t} dt$$

Hence $E[X|Y = y] = 2/(\mu+y)$

Recall that $E[T_k] = k/\mu$ when $T_k \sim \text{Gamma}(k, \mu)$

-
- Hence the best estimate Person 3 can make about the number that Person 1 drew is to observe the magnitude of y that the number Person 2 provided.
 - If y is large number than Person 3 should guess a smaller value for x
 - Otherwise, she/he should guess a larger number.
 - **If the three people play this game many times, Person 3 will guess correctly, on average, the number that Person 1 draws.**

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Exercise 1 – Bitcoin alerts

- Bitcoin trades continuously, and large price movements can occur at unpredictable times. A risk-monitoring system flags an alert whenever Bitcoin's price makes a sudden jump of a certain (large) size. These alerts occur independently and without any predictable pattern.
- It is midnight (00:00), and you are sitting at your desk watching the crypto dashboard. You wonder: what is the probability that the first such alert occurs between 6 am and 6 pm later today?
- Assume the time until the first alert is modeled as an exponential random variable with mean 5 days.

Solution of Exercise 1

- We define the CRV: T = "the time that has elapsed until the 1st alert happens".
- We know that Then T must follow the Exponential distribution but we do not know its mean in one day.
- But we were given its mean for 5 days. Hence $\mu = \frac{1}{5}$ is the rate per day.
- We note that 24 hours can be split into four 6-hour intervals.
- Hence the interval of interest where we are interested the alert event to happen is $[\frac{1}{4}, \frac{3}{4}]$. Consequently, we are interested in the probability of the event $\{\frac{1}{4} \leq X \leq \frac{3}{4}\}$

$$P\left\{\frac{1}{4} \leq T \leq \frac{3}{4}\right\} = P\left\{T \geq \frac{1}{4}\right\} - P\left\{T \leq \frac{3}{4}\right\} = e^{-\frac{1}{5} \times \frac{1}{4}} - e^{-\frac{1}{5} \times \frac{3}{4}} = 0.090521448$$

Exercise 2

- A CRV X has PDF given by the function

$$f(x) = \mu e^{-\mu x}, \text{ if } x \geq 0 \quad \text{and} \quad f(x) = 0, \text{ if } x < 0.$$

- (a) Find the CDF of this random variable? How do we call this random variables?
- (b) Find the rate μ such that $P\{X \leq 1\} = P\{X > 1\}$
- (c) Find the conditional probability: $P\{X > 2 | X > 1\}$

Solution of Exercise 2

- (a) Assume $x \geq 0$. Then the CDF of X is given by

$$\begin{aligned} F(x) &= P\{X \leq x\} = \int_{-\infty}^x f(t)dt = \int_{-\infty}^0 0 dt + \int_0^x \mu e^{-\mu t} dt = \int_0^x \mu e^{-\mu t} dt = \\ &= [-e^{-\mu t}]_{t=0}^x = -e^{-\mu x} - (-e^{-\mu 0}) = -e^{-\mu x} + 1 = 1 - e^{-\mu x} \end{aligned}$$

- For the when $x < 0$ we have $F(x) = P\{X < x\} = 0$.
 - This is usually omitted, but we include it here for completeness.
- This is the CDF of the exponential CRV

Solution of Exercise 2

- (b) We want to compute μ such that $P\{X \leq 1\} = P\{X > 1\}$
- We have $P\{X \leq 1\} = F(1) = 1 - e^{-\mu}$ and $P\{X > 1\} = 1 - P\{X \leq 1\} = e^{-\mu}$
- We set them equal: $1 - e^{-\mu} = e^{-\mu} \Rightarrow 2e^{-\mu} = 1 \Rightarrow e^{-\mu} = \frac{1}{2} \Rightarrow -\mu = \ln\left(\frac{1}{2}\right) \Rightarrow$

$$\Rightarrow -\mu = -\ln(2) \Rightarrow \mu = \ln(2)$$

- (c) we need to find the conditional probability $P\{X > 2|X > 1\} =$

$$= \frac{P(\{X > 2\} \cap \{X > 1\})}{P\{X > 1\}} = \frac{P\{X > 2\}}{1 - P\{X \leq 1\}} = \frac{1 - P\{X \leq 2\}}{1 - P\{X \leq 1\}} = \frac{e^{-2\mu}}{e^{-\mu}} = e^{-\mu} = e^{-\ln(2)} = e^{\ln\left(\frac{1}{2}\right)} = \frac{1}{2}$$

Exercise 3

- Let X be a CRV with CDF defined as follows (we say that X follows the *standard logistic* distribution):

$$F(x) = \frac{1}{1 + e^{-x}}, \forall x \in \mathbb{R}$$

- (a) Show that the CDF is a symmetric function at 0 (you are encouraged to plot the CDF to see how it looks like)

- Hint: any random variable Y is symmetric at α if its CDF satisfies

$$F(\alpha - x) + F(\alpha + x) = 1$$

- (b) Find the PDF of X (some Calculus refreshing activity; not used in the remaining of the exercise)
- (c) Compute the probability: $P\{-1 \leq X \leq 1\}$
- (d) Suppose X represents the deviation of a student's exam score from their average grade so far. The value $X=0$ corresponds to performing exactly at the student's average, while positive and negative values represent better or worse-than-average performance, respectively. A deviation of more than 10 points (in either direction) is considered unusually large. Compute the probability of the unusually large deviation: $P\{|X| > 10\}$.